

Hints & Solutions

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Q1 (4) - Properties of Definite Integrals - I

$$\text{Let } I = \int_0^2 |x^2 + 2x - 3| dx$$

$$= \int_0^1 (x^2 + 2x - 3) dx + \int_1^2 |x^2 + 2x - 3| dx \dots (i)$$

$$\text{We have, } x^2 + 2x - 3 = (x + 3)(x - 1)$$

$$x^2 + 2x - 3 > 0 \text{ for } x < -3 \text{ or } x > 1$$

$$x^2 + 2x - 3 < 0 \text{ for } -3 < x < 1$$

$$\Rightarrow |x^2 + 2x - 3| = \begin{cases} (x^2 + 2x - 3), & x < -3 \text{ or } x > 1 \\ -(x^2 + 2x - 3), & -3 < x < 1 \end{cases}$$

Eq. (i) becomes,

$$I = \int_0^1 -(x^2 + 2x - 3) dx + \int_1^2 (x^2 + 2x - 3) dx$$

$$= -\left[\frac{x^3}{3} + x^2 - 3x\right]_0^1 + \left[\frac{x^3}{3} + x^2 - 3x\right]_1^2 = 4$$

Q2 (2) - Properties of Definite Integrals - I

$$\text{Let } I = \int_0^2 [x^2] dx$$

$$= \int_0^1 [x^2] dx + \int_1^{\sqrt{2}} [x^2] dx + \int_{\sqrt{2}}^{\sqrt{3}} [x^2] dx + \int_{\sqrt{3}}^2 [x^2] dx$$

$$= \int_0^1 0 dx + \int_1^{\sqrt{2}} 1 dx + \int_{\sqrt{2}}^{\sqrt{3}} 2 dx + \int_{\sqrt{3}}^2 3 dx$$

$$= 0 + 1(\sqrt{2} - 1) + 2(\sqrt{3} - \sqrt{2}) + 3(2 - \sqrt{3})$$

$$= 5 - \sqrt{3} - \sqrt{2}$$

Q3 (1) - Properties of Definite Integrals - I

$$\text{Let } I = \int_0^a [x^n] dx$$

Let k be a natural number such that $k \leq a^n < k + 1$

$$\text{then } \int_0^a [x^n] dx = \int_0^1 [x^n] dx + \int_1^{2^{1/n}} [x^n] dx + \int_{2^{1/n}}^{3^{1/n}} [x^n] dx + \dots + \int_{(k-1)^{1/n}}^{k^{1/n}} [x^n] dx + \int_{k^{1/n}}^{(a^n)^{1/n}} x^n dx$$

$$= \int_0^1 0 dx + \int_1^{2^{1/n}} 1 dx + \int_{2^{1/n}}^{3^{1/n}} 2 dx + \dots + \int_{(k-1)^{1/n}}^{k^{1/n}} (k-1) dx + \int_{k^{1/n}}^{(a^n)^{1/n}} k dx$$

$$= 0 + 1 \cdot (2^{1/n} - 1) + 2 \cdot (3^{1/n} - 2^{1/n}) + \dots + (k-1) [k^{1/n} - (k-1)^{1/n}] + k(a - k^{1/n})$$

$$\Rightarrow \sum_{r=1}^k (r-1) [r^{1/n} - (r-1)^{1/n}] + k(a - k^{1/n})$$

Q4 (1) - Properties of Definite Integrals - I

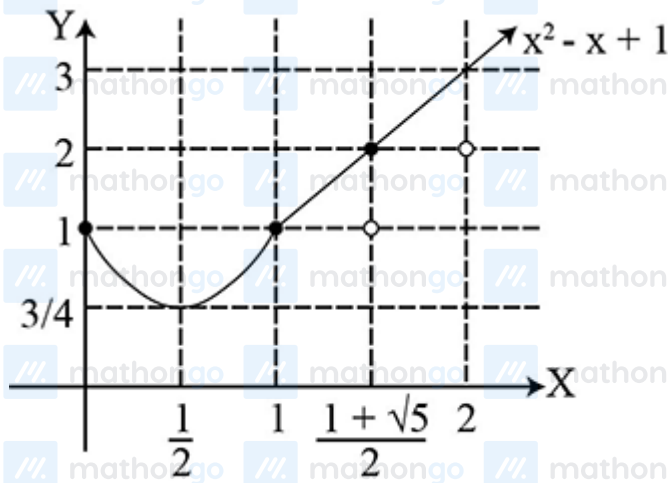
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Here, $f(x) = x^2 - x + 1$ is a non-monotonic function, such problems should be solved by graphical method.

Now, $g(x) = [x^2 - x + 1], \forall x \in [0, 2]$ could be plotted as

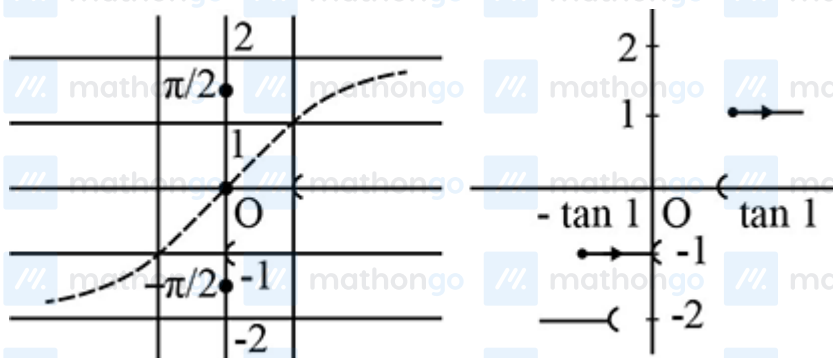


$$\begin{aligned} \therefore I &= \int_0^2 [x^2 - x + 1] dx \\ &= \int_0^1 0 dx + \int_1^{\frac{1+\sqrt{5}}{2}} 1 dx + \int_{\frac{1+\sqrt{5}}{2}}^2 2 dx \\ &= \left(\frac{1+\sqrt{5}}{2} - 1 \right) + 2 \left(2 - \frac{1+\sqrt{5}}{2} \right) = \frac{5-\sqrt{5}}{2} \end{aligned}$$

Q5 (4) - Properties of Definite Integrals - I

Let $I = \int_0^{100} [\tan^{-1} x] dx$

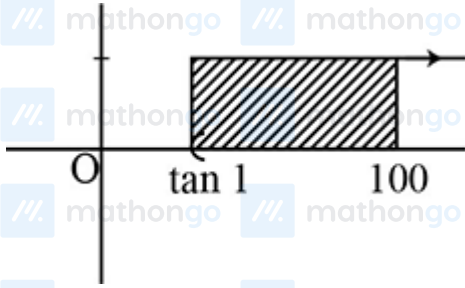
The graph of $\tan^{-1} x$ is shown as



$\therefore \int_0^{100} [\tan^{-1} x] dx$ is shown as

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$$= 1 \times (100 - \tan 1)$$

Q6 (3) - Properties of Definite Integrals - I

Consider the integral, $\int_{1/e}^{\cot x} \frac{dt}{t(1+t^2)}$

On putting $t = \frac{1}{u}$, we get

$$\int_{1/e}^{\cot x} \frac{dt}{t(1+t^2)} = \int_e^{\tan x} \frac{-1/u^2}{\frac{1}{u}(1+\frac{1}{u^2})} du = - \int_e^{\tan x} \frac{u}{1+u^2} du$$

$$\Rightarrow \int_{1/e}^{\cot x} \frac{dt}{t(1+t^2)} = - \int_e^{\tan x} \frac{t}{1+t^2} dt$$

$$= - \left[\int_e^{1/e} \frac{t}{1+t^2} dt + \int_{1/e}^{\tan x} \frac{t}{1+t^2} dt \right]$$

$$\Rightarrow \int_{1/e}^{\cot x} \frac{dt}{t(1+t^2)} = - \int_e^{1/e} \frac{t}{1+t^2} dt - \int_{1/e}^{\tan x} \frac{t}{1+t^2} dt$$

$$\therefore \int_{1/e}^{\tan x} \frac{t}{1+t^2} dt + \int_{1/e}^{\cot x} t(1+t^2) dt$$

$$= - \int_e^{1/e} \frac{t}{1+t^2} dt = - \frac{1}{2} [\log(1+t^2)]^{1/\theta}$$

$$= - \frac{1}{2} \left[\log\left(1 + \frac{1}{e^2}\right) - \log(1+e^2) \right]$$

$$= - \frac{1}{2} [\log(e^2 + 1) - 2 \log e - \log(1 + e^2)]$$

$$= \log e = 1$$

Q7 (1) - Properties of Definite Integrals - I

Since, $\int_1^a [x] f'(x) dx$

$$= \int_1^2 f'(x) dx + \int_2^3 2f'(x) dx + \dots + \int_{[a]}^a [a] f'(x) dx$$

$$= [f(x)]_1^2 + 2[f(x)]_2^3 + \dots + [a][f(x)]_{[a]}^a$$

$$= f(2) - f(1) + 2f(3) - 2f(2) + \dots + [a]f(a) - [a]f([a])$$

$$= [a]f(a) - \{f(1) + f(2) + \dots + f([a])\}$$

Q8 (4) - Properties of Definite Integrals - II

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Given, $f'(x) = f(x)$

$\Rightarrow f(x) = ce^x$ and since $f(0) = 1$

$\therefore 1 = f(0) = c$

$\therefore f(x) = e^x$ and hence $g(x) = x^2 - e^x$

Thus, $\int_0^1 f(x)g(x)dx = \int_0^1 (x^2e^x - e^{2x}) dx$

$= [x^2e^x]_0^1 - 2 \int_0^1 xe^x dx - \left[\frac{e^{2x}}{2}\right]_0^1$

$= (e - 0) - 2 \left\{ [xe^x]_0^1 - [e^x]_0^1 \right\} - \frac{1}{2}(e^2 - 1)$

$= (e - 0) - 2[(e - 0) - (e - 1)] - \frac{1}{2}(e^2 - 1)$

$= e - \frac{1}{2}e^2 - \frac{3}{2}$

Q9 (3) - Properties of Definite Integrals - II

Given, $f(x) = x^4 + |x|$

$I_1 = \int_0^\pi f(\cos x)dx$

$I_2 = \int_0^{\pi/2} f(\sin x)dx$

Clearly, $f(x)$ is an even function.

Hence, $I_1 = \int_0^\pi f[\cos(\pi - x)]dx$

$= \int_0^\pi f(-\cos x)dx$

$= \int_0^\pi f(\cos x)dx$

$\Rightarrow I_1 = 2 \int_0^{\pi/2} f(\cos x)dx$

$= 2 \int_0^{\pi/2} f(\sin x)dx = 2I_2$

$\therefore \frac{I_1}{I_2} = 2$

Q10 (4) - Properties of Definite Integrals - II

Let $I = \int_0^1 x(1-x)^n dx$

On replacing x by $(1-x)$, we get

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$$\begin{aligned}
 I &= \int_0^1 (1-x)x^n dx \\
 &= \int_0^1 (x^n - x^{n+1}) dx \\
 &= \left[\frac{x^{n+1}}{n+1} - \frac{x^{n+2}}{n+2} \right]_0^1 \\
 &= \left[\frac{1}{n+1} - \frac{1}{n+2} \right]
 \end{aligned}$$

Q11 (2) - Properties of Definite Integrals - II

$$\text{Let } I = \int_0^\pi x f(\sin x) dx \dots (i)$$

$$I = \int_0^\pi (\pi - x) f(\sin x) dx \dots (ii)$$

On adding Eqs. (i) and (ii), we get

$$2I = \int_0^\pi \pi f(\sin x) dx$$

$$I = \frac{\pi}{2} \int_0^\pi f(\sin x) dx$$

$$I = \pi \int_0^{\pi/2} f(\sin x) dx$$

$$\Rightarrow \int_0^\pi x f(\sin x) dx = \pi \int_0^{\pi/2} f(\sin x) dx$$

$$\text{But it is } A \int_0^{\pi/2} f(\sin x) dx$$

$$\therefore A = \pi$$

Q12 (2) - Properties of Definite Integrals - II

$$\text{Let } I = \int_{-3\pi/2}^{-\pi/2} \{ (x + \pi)^3 + \cos^2(x + 3\pi) \} dx$$

$$= \int_{-3\pi/2}^{-\pi/2} [(x + \pi)^3 + \cos^2 x] dx \dots (i)$$

$$= \int_{-3\pi/2}^{-\pi/2} \left[\left(-\frac{\pi}{2} - \frac{3\pi}{2} - x + \pi \right)^3 + \cos^2 \left(-\frac{3\pi}{2} - \frac{\pi}{2} - x \right) \right] dx$$

$$= \int_{-3\pi/2}^{-\pi/2} [(-\pi - x)^3 + \cos^2(-2\pi - x)] dx$$

$$= \int_{-3\pi/2}^{-\pi/2} [-(x + \pi)^3 + \cos^2 x] dx \dots (ii)$$

On adding Eqs. (i) and (ii), we get

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$$\begin{aligned}
 2I &= \int_{-3\pi/2}^{-\pi/2} 2 \cos^2 x dx \\
 &= 2 \int_{-3\pi/2}^{-\pi/2} \left(\frac{\cos 2x + 1}{2} \right) dx \\
 &= \int_{-3\pi/2}^{-\pi/2} (\cos 2x + 1) dx \\
 &= \left(\frac{\sin 2x}{2} + x \right)_{-3\pi/2}^{-\pi/2} \\
 &= \left(0 - \frac{\pi}{2} \right) - \left(0 - \frac{3\pi}{2} \right) \\
 &= -\frac{\pi}{2} + \frac{3\pi}{2} \\
 \Rightarrow 2I &= \pi \\
 \therefore 1 &= \frac{\pi}{2}
 \end{aligned}$$

Q13 (2) - Properties of Definite Integrals - II

Let $I = \int_0^\pi [\cot x] dx \dots (i)$

$$= \int_0^\pi [\cot(\pi - x)] dx \quad [\because \int_0^a f(x) dx = \int_0^a f(a - x) dx]$$

$I = \int_0^\pi [-\cot x] dx \dots (ii)$

On adding Eqs. (i) and (ii), we get

$$\begin{aligned}
 2I &= \int_0^\pi \{[\cot x] + [-\cot x]\} dx \\
 &= \int_0^\pi -1 dx \quad \left[\because [x] + [-x] = \begin{cases} -1, x \notin 2 \\ 0, x \in 2 \end{cases} \right] \\
 &= -[x]_0^\pi \\
 \Rightarrow 2I &= -\pi \\
 \therefore I &= -\frac{\pi}{2}
 \end{aligned}$$

Q14 (3) - Properties of Definite Integrals - II

Let $I = \int_0^\pi e^{\cos^2 x} \cos^3(2n+1)x dx$

$$\begin{aligned}
 &= \int_0^\pi e^{\cos^2(\pi-x)} \cos^3(2n+1)(\pi-x) dx \\
 &= \int_0^\pi e^{\cos^2 x} \cos^3\{(2n+1)\pi - (2n+1)x\} dx \\
 &= \int_0^\pi -e^{\cos^2 x} \cos^3(2n+1)x dx
 \end{aligned}$$

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$$= - \int_0^\pi e^{\cos^2 x} \cos^3(2n+1)x dx$$

$$\Rightarrow I = -I \Rightarrow 2I = 0$$

$$\therefore I = 0$$

Q15 (1) - Properties of Definite Integrals - II

$$\text{Let } I = \int_{-1/2}^{1/2} \left[x + \log\left(\frac{1+x}{1-x}\right) \right] dx$$

$$= \int_{-1/2}^{1/2} x dx + \int_{-1/2}^{1/2} \log\left(\frac{1+x}{1-x}\right) dx$$

$$= \int_{-1/2}^0 x dx + \int_0^{1/2} x dx + \int_{-1/2}^{1/2} \log\left(\frac{1+x}{1-x}\right) dx$$

$$= \int_{-1/2}^0 x dx + \int_0^{1/2} 0 dx + 0 \left[\because \log\left(\frac{1+x}{1-x}\right) \text{ is an odd function.} \right]$$

$$= \left[-x^2 \right]_{-1/2}^0$$

$$= - \left[0 + \frac{1}{2} \right] = -\frac{1}{2}$$

Q16 (2) - Properties of Definite Integrals - II

$$\text{Let } I = \int_{-\pi/2}^{\pi/2} \left[x^2 + \log\left(\frac{\pi+x}{\pi-x}\right) \right] \cos x dx$$

$$= \int_{-\pi/2}^{\pi/2} x^2 \cos x dx + \int_{-\pi/2}^{\pi/2} \log\left(\frac{\pi+x}{\pi-x}\right) \cos x dx$$

$$= \int_{-\pi/2}^{\pi/2} x^2 \cos x dx + 0 \left[\because \log\left(\frac{\pi+x}{\pi-x}\right) \cos x \text{ is an odd function.} \right]$$

$$= \int_{-\pi/2}^{\pi/2} x^2 \cos x dx$$

$$= \left[x^2 \sin x \right]_{-\pi/2}^{\pi/2} - \int_{-\pi/2}^{\pi/2} 2x \cdot \sin x dx$$

$$= \left[x^2 \sin x \right]_{-\pi/2}^{\pi/2} - 2 \left[-x \cos x - \int -\cos x dx \right]_{-\pi/2}^{\pi/2}$$

$$= \left[x^2 \sin x \right]_{-\pi/2}^{\pi/2} - 2 \left[-x \cos x + \sin x \right]_{-\pi/2}^{\pi/2}$$

$$= \left[x^2 \sin x + 2x \cos x - 2 \sin x \right]_{-\pi/2}^{\pi/2}$$

$$= \left(\frac{\pi^2}{4} + 0 - 2 \right) - \left(-\frac{\pi^2}{4} + 0 + 2 \right)$$

$$= \frac{\pi^2}{4} - 2 + \frac{\pi^2}{4} - 2$$

$$= \frac{2\pi^2}{4} - 4$$

$$= \frac{\pi^2}{2} - 4$$

Q17 (1) - Periodic Function and Leibnitz Theorem

$$\text{We have, } \lim_{x \rightarrow 1} \int_4^{f(x)} \frac{2t}{x-1} dt$$

$$= \lim_{x \rightarrow 1} \frac{2f(x)f'(x)}{1} = 2f(1) \cdot f'(1)$$

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$$= 2 \times 4f'(1)$$

$$= 8f'(1)$$

Q18 (2) - Periodic Function and Leibnitz Theorem

$$\text{We have, } \int_0^{n\pi+v} \sqrt{\frac{1+\cos 2x}{2}} dx$$

$$= \int_0^{n\pi+v} \sqrt{\cos^2 x} dx$$

$$= \int_0^{n\pi+v} |\cos x| dx$$

$$= \int_0^v |\cos x| dx + \int_v^{v+n\pi} |\cos x| dx$$

$$= -\int_0^v \cos x dx + n \int_0^\pi |\cos x| dx$$

$$= [-\sin x]_0^v + n \left[\int_0^{\pi/2} \cos x dx + \int_{\pi/2}^\pi -\cos x dx \right]$$

$$= [-\sin x]_0^v + n \int_0^{\pi/2} \cos x dx - n \int_{\pi/2}^\pi \cos x dx$$

$$= [-\sin x]_0^v + n[\sin x]_0^{\pi/2} - n[\sin x]_{\pi/2}^\pi$$

$$= -\sin v + n + n = 2n - \sin v$$

Q19 (2) - Periodic Function and Leibnitz Theorem

$$\text{Given, } f(x) = \int_0^{\sin^2 x} \sin^{-1} \sqrt{t} dt + \int_0^{\cos^2 x} \cos^{-1} \sqrt{t} dt$$

$$\Rightarrow f'(x) = \sin^{-1} \sqrt{\sin^2 x} \cdot 2 \sin x \cos x + \cos^{-1} \sqrt{\cos^2 x} \cdot 2 \cos x (-\sin x)$$

$$= 2x \sin x \cos x - 2x \sin x \cos x$$

$$= 0$$

$\Rightarrow f(x)$ is a constant function and is independent of x . So, let's select

$$x = \frac{\pi}{4} \quad \left(\because x \in \left[0, \frac{\pi}{2}\right] \right)$$

$$\therefore f(x) = \int_0^{1/2} \sin^{-1} \sqrt{t} dt + \int_0^{1/2} \cos^{-1} \sqrt{t} dt$$

$$= \int_0^{1/2} (\sin^{-1} \sqrt{t} + \cos^{-1} \sqrt{t}) dt$$

$$= \int_0^{1/2} \frac{\pi}{2} dt$$

$$= \frac{\pi}{2} [t]_0^{1/2} = \frac{\pi}{2} \times \frac{1}{2} = \frac{\pi}{4}$$

Q20 (4) - Periodic Function and Leibnitz Theorem

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$$\begin{aligned}
 &\text{We have, } \lim_{x \rightarrow 2} \int_6^{f(x)} \frac{4t^3}{x-2} dt \\
 &= \lim_{x \rightarrow 2} \frac{\int_6^{f(x)} 4t^3 dt}{x-2} = \lim_{x \rightarrow 2} \frac{4[f(x)]^3 f'(x)}{1} \\
 &= 4 \lim_{x \rightarrow 2} [f(x)]^3 f'(x) = 4[f(2)]^3 f'(2) \\
 &= 4(6)^3 \times \frac{1}{48} \quad \left[\begin{array}{l} \because f(2) = 6 \\ f'(2) = \frac{1}{48} \end{array} \right] \\
 &= 18
 \end{aligned}$$

Q21 (3) - Periodic Function and Leibnitz Theorem

$$\begin{aligned}
 &\text{We have, } F(x) = \int_0^x f(t) dt \\
 &\Rightarrow F'(x) = f(x) \quad \dots \text{ (i) [Leibnitz rule]} \\
 &\text{and } F(x^2) = x^2(1+x) = x^2 + x^3 \quad [\text{given}] \\
 &\Rightarrow F(x) = x + x^{3/2} \\
 &\Rightarrow F'(x) = 1 + \frac{3}{2}x^{1/2} \\
 &\Rightarrow f(x) = 1 + \frac{3}{2}x^{1/2} \quad [\text{from Eq.(i)}] \\
 &\Rightarrow f(4) = 1 + \frac{3}{2} \times (4)^{1/2} \\
 &= 1 + \frac{3}{2} \times 2 \\
 &\therefore f(4) = 4
 \end{aligned}$$

Q22 (3) - Periodic Function and Leibnitz Theorem

$$\begin{aligned}
 &\text{We have, } I_1 = \int_3^{3+3T} f(2x) dx \\
 &\text{Put } 2x = t \Rightarrow dx = \frac{dt}{2} \\
 &\therefore I_1 = \frac{1}{2} \int_6^{6+6T} f(t) dt \\
 &= \frac{1}{2} \times 6 \int_0^T f(t) dt \\
 &= 3I \quad \left[\because I = \int_0^T f(x) dx \right]
 \end{aligned}$$

Q23 (2) - Periodic Function and Leibnitz Theorem

$$\begin{aligned}
 &\text{Given, } f(x) = \int_{x^2}^{x^2+1} e^{-t^2} dt \\
 &\text{On differentiating both sides by using Leibnitz rule, we get} \\
 &f'(x) = e^{-(x^2+1)^2} \cdot 2x - e^{-(x^2)^2} \cdot 2x \\
 &f'(x) = 2xe^{-(x^4+2x^2+1)} (1 - e^{2x^2+1})
 \end{aligned}$$

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[where, $e^{2x^2+1} > 1, \forall x$ and $e^{-(x^4+2x^2+1)} > 0, \forall x$]

$$\therefore f'(x) \geq 0$$

which shows $2x \leq 0$

$$\Rightarrow x \leq 0$$

Q24 (3) - Periodic Function and Leibnitz Theorem

$$\text{Given, } \int_0^x \sqrt{1 - [f'(t)]^2} dt = \int_0^x f(t) dt, 0 \leq x \leq 1$$

On differentiating w.r.t. x by using Leibnitz rule, we get

$$\sqrt{1 - [f'(x)]^2} = f(x)$$

$$\Rightarrow f'(x) = \pm \sqrt{1 - [f(x)]^2}$$

$$\Rightarrow \int \frac{f'(x)}{\sqrt{1 - [f(x)]^2}} dx = \pm \int dx$$

$$\Rightarrow \sin^{-1}[f(x)] = \pm x + C$$

$$\text{Put } x = 0 \Rightarrow \sin^{-1}[f(0)] = C$$

$$\Rightarrow C = \sin^{-1}(0) = 0$$

$$\therefore f(x) = \pm \sin x$$

$$\text{But } f(x) \geq 0, \forall x \in [0, 1]$$

$$\therefore f(x) = \sin x$$

As we know that

$$\sin x < x, \forall x > 0$$

$$\sin\left(\frac{1}{2}\right) < \frac{1}{2} \text{ and } \sin\left(\frac{1}{3}\right) < \frac{1}{3}$$

$$f\left(\frac{1}{2}\right) < \frac{1}{2} \text{ and } f\left(\frac{1}{3}\right) < \frac{1}{3}$$

Q25 (2) - Periodic Function and Leibnitz Theorem

$$\text{We have, } e^{-x} f(x) = 2t \int_0^x \sqrt{t^4 + 1} dt, x \in (-1, 1)$$

On differentiating w.r.t. x using Leibnitz rule, we get

$$e^{-x} [f'(x) - f(x)] = \sqrt{x^4 + 1}$$

$$f'(x) = f(x) + \sqrt{x^4 + 1} e^x$$

$$\therefore f^{-1} \text{ is inverse of } f$$

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$$\therefore f^{-1}[f(x)] = x$$

$$\Rightarrow (f^{-1})'[f(x)]f'(x) = 1$$

$$\Rightarrow (f^{-1})'[f(x)] = \frac{1}{f'(x)} = \frac{1}{f(x) + \sqrt{x^4 + 1}e^x}$$

$$\text{At } x = 0, \quad f(x) = 2$$

$$\therefore (f^{-1})'(2) = \frac{1}{2+1} = \frac{1}{3}$$

Q26 (2) - Estimation and Gamma Function

We have,

$$\sin x < x \text{ for } 0 < x < 1$$

$$\Rightarrow \frac{\sin x}{\sqrt{x}} < \frac{x}{\sqrt{x}} \text{ for } 0 < x < 1$$

$$\Rightarrow \int_0^1 \frac{\sin x}{\sqrt{x}} dx < \int_0^1 \sqrt{x} dx$$

$$\Rightarrow 1 < \frac{2}{3} [x^{3/2}]_0^1$$

$$\Rightarrow 1 < \frac{2}{3}$$

$$\text{Also, } \cos x < 1 \text{ for all } x \in (0, 1)$$

$$\frac{\cos x}{\sqrt{x}} < \frac{1}{\sqrt{x}} \text{ for all } x \in (0, 1)$$

$$\Rightarrow \int_0^1 \frac{\cos x}{\sqrt{x}} dx < \int_0^1 \frac{1}{\sqrt{x}} dx$$

$$\Rightarrow J < 2 [x^{1/2}]_0^1$$

$$\therefore J < 2$$

Q27 (3) - Estimation and Gamma Function

$$\text{Given, } l(m, n) = \int_0^1 t^m (1+t)^n dt$$

$$= \left[(1+t)^n \cdot \frac{t^{m+1}}{m+1} \right]_0^1 - \int_0^1 \left[n(1+t)^{n-1} \cdot \frac{t^{m+1}}{m+1} \right] dt$$

$$= \frac{2^n}{m+1} - \frac{n}{m+1} \cdot \int_0^1 (1+t)^{n-1} \cdot t^{m+1} dt$$

$$\therefore l(m, n) = \frac{2^n}{m+1} - \frac{n}{m+1} \cdot l(m+1, n-1)$$

Q28 (3) - Estimation and Gamma Function

$$\text{To find } I = \left| \int_{10}^{19} \frac{\sin x}{1+x^8} dx \right| \leq \int_{10}^{19} \left| \frac{\sin x}{1+x^8} \right| dx \dots (i)$$

$$\left[\because \int_a^b f(x) dx = \int_a^b |f(x)| dx \right]$$

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Since, $|\sin x| \leq 1$ for $x \geq 10$

The inequality $\left| \frac{\sin x}{1+x^8} \right| \leq \frac{1}{1+x^8}$... (ii)

Also, $10 \leq x \leq 19$

$$\Rightarrow 1 + x^8 > 10^8$$

$$\Rightarrow \frac{1}{1+x^8} < \frac{1}{10^8}$$

Or $\frac{1}{1+x^8} < 10^{-8}$... (iii)

From Eqs. (ii) and (iii), we get

$\left| \frac{\sin x}{1+x^8} \right| < 10^{-8}$ is fulfilled.

$$\int_{10}^{19} \frac{\sin x}{1+x^8} dx < \int_{10}^{19} 10^{-8} dx$$

$$\therefore \int_{10}^{19} \frac{\sin x}{1+x^8} dx < (19-10)10^{-8} < 10^{-7}$$

Q29 (1) - Limit as a Sum

Given,

$$\begin{aligned} S_n &= \frac{n}{(n+1)(n+2)} + \frac{n}{(n+2)(n+3)} + \frac{n}{(n+3)(n+4)} + \dots + \frac{1}{6n^2} \\ &= \sum_{r=1}^n \frac{n}{(n+r)(n+2r)} \\ &= \sum_{r=1}^n \frac{n}{n^2 \left(1+\frac{r}{n}\right) \left(1+\frac{2r}{n}\right)} \\ &= \frac{1}{n} \sum_{r=1}^n \frac{1}{\left(1+\frac{r}{n}\right) \left(1+\frac{2r}{n}\right)} \end{aligned}$$

$$\text{Now, } \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[\frac{1}{n} \sum_{r=1}^n \frac{1}{\left(1+\frac{r}{n}\right) \left(1+\frac{2r}{n}\right)} \right]$$

For upper limit,

$$\lim_{n \rightarrow \infty} \frac{r}{n} = \lim_{n \rightarrow \infty} \frac{n}{n} = 1$$

For lower limit,

$$\lim_{n \rightarrow \infty} \frac{r}{n} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$= \int_0^1 \frac{1}{(1+x)(1+2x)} dx$$

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Now, $\frac{1}{(1+x)(1+2x)} = \frac{A}{1+x} + \frac{B}{1+2x}$

Put $1 = A(1+2x) + B(1+x)$

$\Rightarrow x = -1/2$

Put $x = -1 \Rightarrow I = \frac{B}{2} \Rightarrow B = 2$

$\Rightarrow 1 = -A$

$\Rightarrow A = -1$

$\therefore \int_0^1 \frac{dx}{(1+x)(1+2x)} = - \int_0^1 \frac{dx}{1+x} + 2 \int_0^1 \frac{dx}{1+2x}$

$= [-\log(1+x)]_0^1 + [\log(1+2x)]_0^1$

$= -\log 2 + (\log 3 - 0)$

$= -\log 2 + \log 3 = \log(3/2)$

Q30 (2) - Limit as a Sum

We have, $\lim_{n \rightarrow \infty} \frac{1^p + 2^p + \dots + n^p}{n^{p+1}}$

$\Rightarrow \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{r^p}{n^{p+1}} = \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{r}{n}\right)^p \cdot \frac{1}{n}$

Put $\frac{r}{n} = x$

For lower limit, $\lim_{n \rightarrow \infty} \frac{r}{n} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$

For upper limit, $\lim_{n \rightarrow \infty} \frac{r}{n} = \lim_{n \rightarrow \infty} \frac{n}{n} = 1$

$= \int_0^1 x^p dx = \left[\frac{x^{p+1}}{p+1} \right]_0^1$

$= \left[\frac{1}{p+1} - 0 \right] = \frac{1}{p+1}$

Q31 (2) - Limit as a Sum

We have, $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} e^{r/n}$

On replacing $\frac{r}{n} = x \Rightarrow \frac{1}{n} = dx$

and $\lim_{n \rightarrow \infty} \sum_{r=1}^n = \int$

For lower limit,

$\lim_{n \rightarrow \infty} \frac{r}{n} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$

For upper limit,

$\lim_{n \rightarrow \infty} \frac{r}{n} = \lim_{n \rightarrow \infty} \frac{n}{n} = 1$

$\int_0^1 e^x dx = [e^x]_0^1 = (e - 1)$

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Q32 (4) - Limit as a Sum

We have,

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n^2} \sec^2 \frac{1}{n^2} + \frac{2}{n^2} \sec^2 \frac{4}{n^2} + \dots + \frac{1}{n} \sec^2 1 \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{n^2} \sec^2 \frac{1}{n^2} + \frac{2}{n^2} \sec^2 \frac{4}{n^2} + \dots + \frac{1}{n} \sec^2 \frac{n^2}{n^2} \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{r}{n^2} \sec^2 \frac{r^2}{n^2} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left[\left(\frac{r}{n} \right) \sec^2 \left(\frac{r}{n} \right)^2 \right]$$

On replacing $\frac{r}{n} = x$

$$\Rightarrow 1/n = dx$$

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n = \int$$

$$\text{For lower limit, } \lim_{n \rightarrow \infty} \frac{r}{n} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\text{For upper limit, } \lim_{n \rightarrow \infty} \frac{r}{n} = \lim_{n \rightarrow \infty} \frac{n}{n} = 1$$

$$\therefore \int_0^1 x \sec^2 x^2 dx$$

$$\text{Put } x^2 = t \Rightarrow 2x dx = dt$$

$$\int_0^1 \sec^2 t \frac{dt}{2} = \frac{1}{2} \int_0^1 \sec^2 t dt$$

$$= \frac{1}{2} [\tan t]_0^1$$

$$= \frac{1}{2} [\tan(1) - \tan(0)]$$

$$= \frac{1}{2} \tan 1$$

Q33 (3) - Limit as a Sum

We have,

$$\lim_{n \rightarrow \infty} \sum_{r=1}^{4n} \frac{\sqrt{n}}{\sqrt{r}(3\sqrt{r}+4\sqrt{n})^2}$$

$$\therefore T_r = \frac{1}{\sqrt{\frac{r}{n}n} \left(3\sqrt{\frac{r}{n}+4} \right)^2}$$

$$S = \frac{1}{n} \sum_{r=1}^{4n} \frac{1}{\left(3\sqrt{\frac{r}{n}+4} \right)^2 \sqrt{\frac{r}{n}}}$$

$$= \int_0^4 \frac{dx}{\sqrt{x}(3\sqrt{x}+4)^2}$$

$$\text{Put } 3\sqrt{x}+4 = t$$

$$\Rightarrow \frac{3}{2} \cdot \frac{1}{\sqrt{x}} dx = dt$$

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$$\begin{aligned}\therefore S &= \frac{2}{3} \int_4^{10} \frac{dt}{t^2} \\ &= \frac{2}{3} \left[\frac{1}{t} \right]_4^{10} \\ &= \frac{2}{3} \left[\frac{1}{4} - \frac{1}{10} \right] \\ &= \frac{2}{3} \cdot \frac{6}{40} = \frac{1}{10}\end{aligned}$$

Q34 (4) - Area under Curves

We have,

$$\int_1^b f(x) dx = \sqrt{b^2 + 1} - \sqrt{2}$$

On differentiating w.r.t. b , we get

$$f(b) = \frac{b}{\sqrt{b^2 + 1}}$$

$$\Rightarrow f(x) = \frac{x}{\sqrt{x^2 + 1}}$$

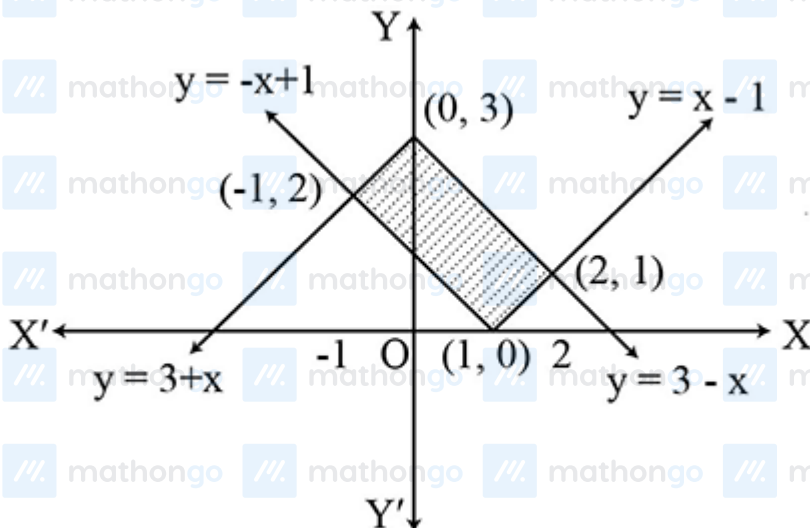
Q35 (4) - Area under Curves

Given curves

$$y = |x - 1| \dots (i)$$

$$\text{and } y = 3 - |x| \dots (ii)$$

The graph of the curves is given below

 $\therefore$ The required area

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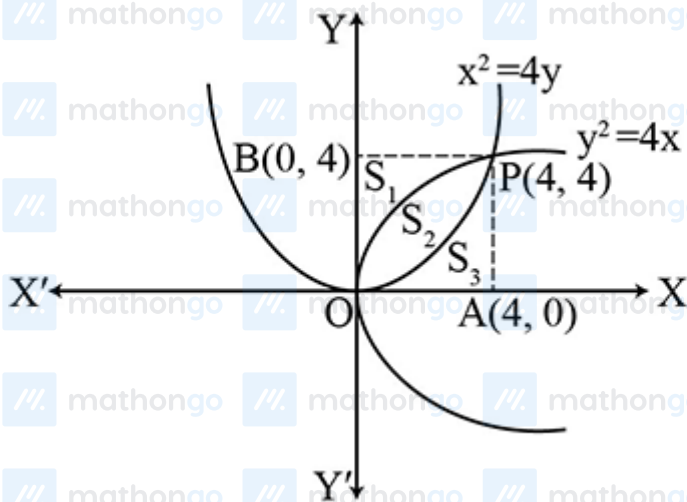
$$\begin{aligned}
 &= \int_{-1}^0 \{(3+x) - (-x+1)\} + \int_0^1 \{(3-x) - (-x+1)\} dx + \int_1^2 \{(3-x) - (x-1)\} dx \\
 &= \int_{-1}^0 (2+2x) dx + \int_0^1 2 dx + \int_1^2 (4-2x) dx \\
 &= [2x + x^2]_{-1}^0 + [2x]_0^1 + [4x - x^2]_1^2 \\
 &= 4 \text{ sq units}
 \end{aligned}$$

Q36 (4) - Area under Curves

Given curves are

$$y^2 = 4x \dots (i)$$

$$x^2 = 4y \dots (ii)$$

Clearly, $y^2 = 4x$ and $x^2 = 4y$ intersect at $(4, 4)$.

$$\begin{aligned}
 \therefore S_2 + S_3 &= \int_0^4 \sqrt{4x} dx \\
 \Rightarrow S_2 + S_3 &= 2 \int_0^4 \sqrt{x} dx = \frac{4}{3} \left[x^{3/2} \right]_0^4 = \frac{32}{3} \text{ sq units} \\
 \text{and } S_3 &= \int_0^4 \frac{x^2}{4} dy = \frac{1}{4} \left[\frac{y^3}{3} \right]_0^4 = \frac{16}{3} \text{ sq units}
 \end{aligned}$$

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Thus, we have

$$S_2 + S_3 = \frac{32}{3} \text{ and } S_3 = \frac{16}{3}$$

$$\Rightarrow S_2 = \frac{16}{3} \text{ sq units}$$

$$\therefore S_1 + S_2 + S_3 = \text{Area of square OAPB}$$

$$= 4 \times 4 = 16 \text{ sq units}$$

$$\therefore S_1 = \frac{16}{3} \text{ sq units} \left[\because S_2 = S_3 = \frac{16}{3} \text{ sq units} \right]$$

$$\text{Hence, } S_1 : S_2 : S_3 = 1 : 1 : 1$$

Q37 (1) - Area under Curves

Given curves are

$$x = -2y^2 \dots (i)$$

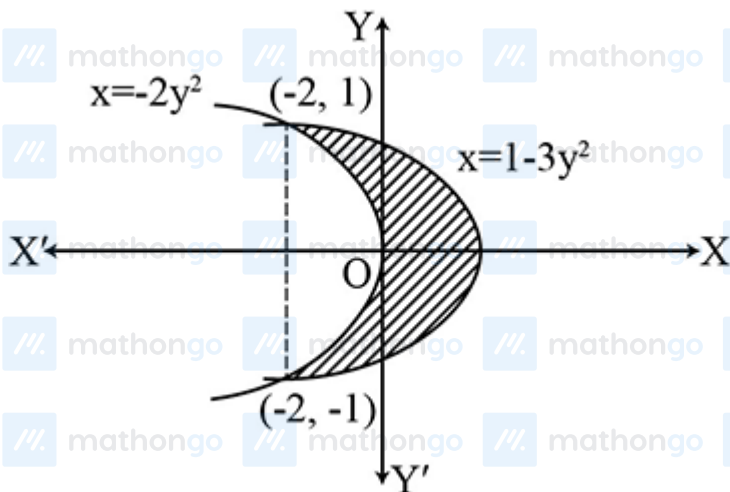
$$\text{and } x = 1 - 3y^2 \dots (ii)$$

To find points of intersection, we solve Eqs. (i) and (ii) simultaneously, we get

$$-2y^2 = 1 - 3y^2 \Rightarrow y^2 = 1$$

$$\Rightarrow y \pm 1$$

$\therefore$ Points of intersection are $(-2, 1)$ and $(-2, -1)$ Clearly, curve $x = -2y^2$ represents left hand parabola with vertex $(0, 0)$ and curve $x = 1 - 3y^2$ represents left hand parabola with vertex $(1, 0)$ as shown in the figure.



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∴ Required area

$$\begin{aligned}
 &= 2 \left[\int_0^1 x \cdot dy \text{ (for } x = 1 - 3y^2) - \int_0^1 x dy \text{ (for } x = -2y^2) \right] \\
 &= 2 \left[\int_0^1 (1 - 3y^2) dy - \int_0^1 (-2y^2) dy \right] \\
 &= 2 \left[\int_0^1 (1 - 3y^2 + 2y^2) dy \right] \\
 &= 2 \left[\int_0^1 (1 - y^2) dy \right] = 2 \left[y - \frac{y^3}{3} \right]_0^1 \\
 &= 2 \left[\left(1 - \frac{1}{3}\right) - 0 \right] = \frac{4}{3} \text{ sq units}
 \end{aligned}$$

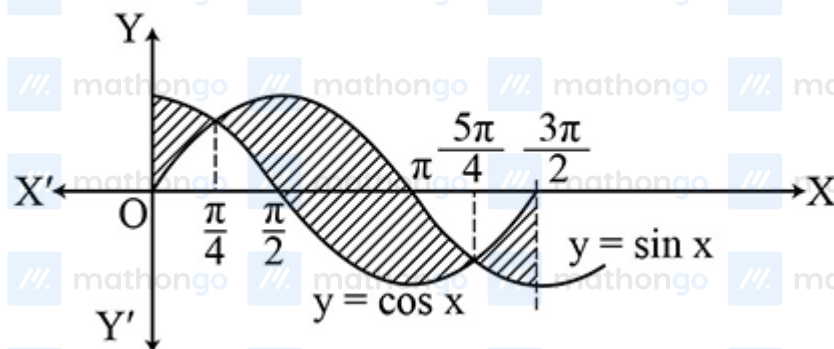
Q38 (1) - Area under Curves

Given curves are

$$y = \cos x \dots (i)$$

$$\text{and } y = \sin x \dots (ii)$$

The graph of curves (i) and (ii) is shown below.



∴ Required area

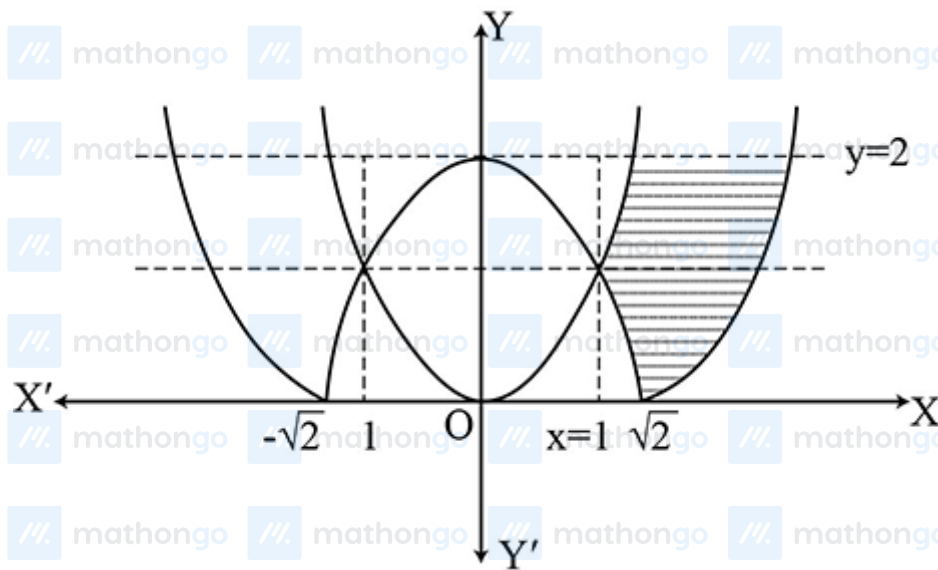
$$\begin{aligned}
 &= \int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{5\pi/4} (\sin x - \cos x) dx + \int_{5\pi/4}^{3\pi/2} (\cos x - \sin x) dx \\
 &= [\sin x + \cos x]_0^{\pi/4} + [-\cos x - \sin x]_{\pi/4}^{5\pi/4} + [\sin x + \cos x]_{5\pi/4}^{3\pi/2} \\
 &= (4\sqrt{2} - 2) \text{ sq units}
 \end{aligned}$$

Q39 (1) - Area under Curves

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The region bounded by given curves on the right side of $x = 1$ is shown.



Thus, the required area

$$\begin{aligned}
 &= \int_1^{\sqrt{2}} \{x^2 - (2 - x^2)\} dx + \int_{\sqrt{2}}^2 \{4 - x^2\} dx \\
 &= \int_1^{\sqrt{2}} (2x^2 - 2) dx + \int_{\sqrt{2}}^2 (4 - x^2) dx \\
 &= \left[2\frac{x^3}{3} - 2x \right]_1^{\sqrt{2}} + \left[4x - \frac{x^3}{3} \right]_{\sqrt{2}}^2 \\
 &= \left(\frac{4\sqrt{2}}{3} - 2\sqrt{2} \right) - \left(\frac{2}{3} - 2 \right) + \left(8 - \frac{8}{3} \right) - \left(4\sqrt{2} - \frac{2\sqrt{2}}{3} \right) \\
 &= \left(-4\sqrt{2} + \frac{20}{3} \right) = \left(\frac{20 - 12\sqrt{2}}{3} \right) \text{ sq units}
 \end{aligned}$$

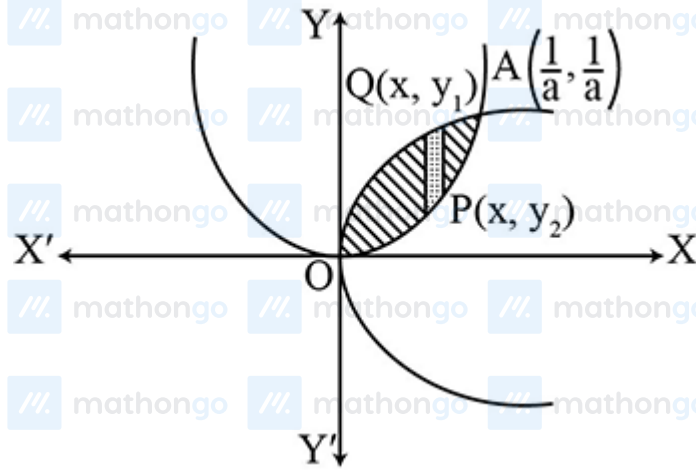
Q40 (1) - Area under Curves

Given curves are $y = ax^2 \dots(i)$

and $x = ay^2 \dots(ii)$

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We have, $\int_0^{1/a} (y_2 - y_1) dx = 1$

$$\Rightarrow \int_0^{1/a} \left(\sqrt{\frac{x}{a}} - ax^2 \right) dx = 1$$

$$\Rightarrow \left[\frac{2}{3\sqrt{a}} x^{3/2} - \frac{ax^3}{3} \right]_0^{1/a} = 1$$

$$\Rightarrow \frac{2}{3\sqrt{a}} \times \frac{1}{a^{3/2}} - \frac{a}{3} \times \frac{1}{a^3} = 1$$

$$\Rightarrow \frac{2}{3a^2} - \frac{1}{3a^2} = 1$$

$$\Rightarrow \frac{1}{3a^2} = 1 \Rightarrow a^2 = \frac{1}{3}$$

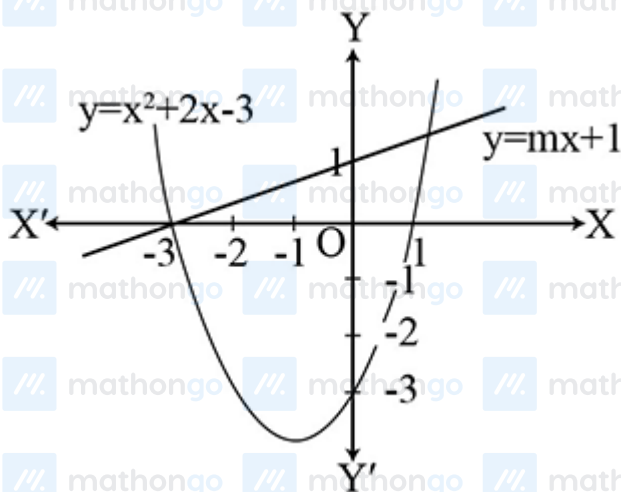
$$\Rightarrow a = \frac{1}{\sqrt{3}} \quad [\because a > 0]$$

Q41 (3) - Area under Curves

Let x_1, x_2 be the roots of equation

$$x^2 + 2x - 3 = mx + 1$$

$$\Rightarrow x^2 + (2 - m)x - 4 = 0$$



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Now, $x_1 + x_2 = m - 2$

and $x_1 \cdot x_2 = -4$

$\therefore$ Required area (A) = $\int_{x_1}^{x_2} [(mx + 1) - (x^2 + 2x - 3)] dx$

= $\left[(m-2) \frac{x^2}{2} - \frac{x^3}{3} + 4x \right]_{x_1}^{x_2}$

= $(m-2) \left(\frac{x_2^2 - x_1^2}{2} \right) - \frac{1}{3} (x_2^3 - x_1^3) + 4(x_2 - x_1)$

= $(x_2 - x_1) \left[\frac{(m-2)^2}{2} - \frac{1}{3} \{ (x_2 + x_1)^2 - x_1 x_2 \} + 4 \right]$

= $\sqrt{(x_2 + x_1)^2 - 4x_1 x_2} \left[\frac{(m-2)^2}{2} - \frac{1}{3} \{ (m-2)^2 + 4 \} + 4 \right]$

= $\frac{\sqrt{(m-2)^2 + 16}}{6} \left[\frac{1}{6} (m-2)^2 + \frac{16}{3} \right]$

= $\frac{[(m-2)^2 + 16]^{3/2}}{6}$

$\Rightarrow$ A is minimum, if $m = 2$.

Hence, $A_{\min} = \frac{32}{3}$ sq units

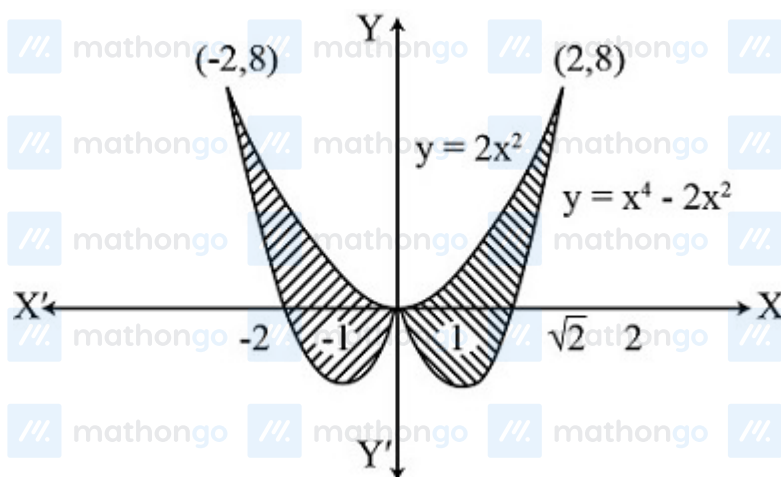
Q42 (2) - Area under Curves

Given, $y = 2x^2$... (i)

and $y = x^4 - 2x^2$... (ii)

Clearly, the points of intersection of curves (i) and (ii) are (0, 0), (2, 8) and (-2, 8)

The graph curves (i) and (ii) could be plotted



$\therefore$ Required area = $2 \left[\int_0^2 (2x^2 - x^4 + 2x^2) dx \right]$

= $2 \left[\frac{2x^3}{3} - \frac{x^5}{5} + \frac{2x^3}{3} \right]_0^2 = 2 \left[\frac{4}{3}x^3 - \frac{x^5}{5} \right]_0^2$

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$$= 2 \left(\frac{32}{3} - \frac{32}{5} \right) = 2 \left[\frac{64}{15} \right]$$

$$= \frac{128}{15} \text{ sq units}$$

Q43 (1) - Area under Curves

As we know,

$$[-\sin^2 x] = 0 \text{ or } -1$$

But $\sec^{-1}(0)$ is not defined.

$$\Rightarrow \sec^{-1}[-\sin^2 x] = \sec^{-1}(-1) = \pi$$

Thus, to find area bounded between

$$y = x^2 \text{ and } y = \pi$$

i.e. when $x^2 = \pi$ or $(x = -\sqrt{\pi} \text{ to } x = \sqrt{\pi})$

$$\text{Area} = \int_{-\sqrt{\pi}}^{\sqrt{\pi}} (\pi - x^2) dx = \left[\pi x - \frac{x^3}{3} \right]_{-\sqrt{\pi}}^{\sqrt{\pi}}$$

$$= \pi(\sqrt{\pi} + \sqrt{\pi}) - \frac{1}{3}(\pi\sqrt{\pi} + \pi\sqrt{\pi})$$

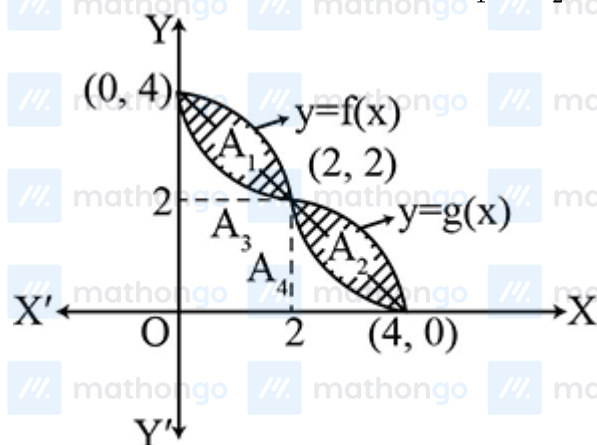
$$= \frac{4\pi}{3}\sqrt{\pi} \text{ sq units}$$

Q44 (3) - Area under Curves

$$\text{Given, } \int_0^4 f(x) dx - \int_0^4 g(x) dx = 10$$

$$\Rightarrow (A_1 + A_3 + A_4) - (A_2 + A_3 + A_4) = 10$$

$$A_1 - A_2 = 10 \quad \dots (i)$$



$$\text{and } \int_2^4 g(x) dx - \int_2^4 f(x) dx = 5$$

$$(A_2 + A_4) - A_4 = 5$$

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$$A_2 = 5 \dots (ii)$$

On adding Eqs. (i) and (ii), we get

$$A_1 = 15 \text{ sq units}$$

Q45 (4) - Area under Curves

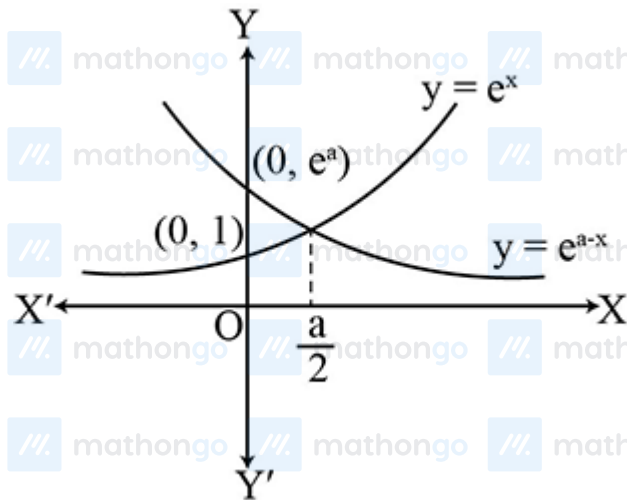
Given, $C_1 : y = e^x \dots (i)$

and $C_2 : y = e^{a-x} \dots (ii)$

On solving Eqs. (i) and (ii), we get

$$e^{2x} = e^a$$

$$\Rightarrow x = \frac{a}{2}$$



Now, $S = \int_0^{a/2} (e^a \cdot e^{-x} - e^x) dx$

$$= [- (e^a \cdot e^{-x} + e^x)]_0^{a/2}$$

$$= (e^a + 1) - (e^{a/2} + e^{a/2})$$

$$= e^a - 2e^{a/2} + 1$$

$$= (e^{a/2} - 1)^2$$

Then,

$$\frac{S}{a^2} = \left(\frac{e^{a/2} - 1}{a} \right)^2 = \frac{1}{4} \left(\frac{e^{a/2} - 1}{a/2} \right)^2$$

$$\lim_{a \rightarrow 0} \frac{S}{a^2} = \frac{1}{4}$$

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